Module 0 Homework 1 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:11

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Fill in the blanks:

$1 - \frac{1}{2}$ = ___,

$\frac{1}{2} - \frac{1}{3}$ = ___, and

$\frac{1}{3} - \frac{1}{4}$ = ___.

- $\frac{1}{2}; \frac{1}{3}; \frac{1}{4}$
- $\frac{1}{2}; \frac{1}{6}; \frac{1}{10}$
- $\frac{1}{2}; \frac{1}{6}; \frac{1}{12}$
- $1; \frac{1}{2}; \frac{1}{3}$
- $1; 1; 1$

2

Review Later

Fill in the blanks:

$7 \times 7 - 8 \times 6$ = ___,

$7 \times 7 - 9 \times 5$ = ___, and

$7 \times 7 - 10 \times 4$ = ___.

- $1; 2; 3$
- $1; 3; 5$
- $1; 4; 7$
- $1; 4; 9$
- $1; 5; 9$

3

Review Later

Evaluate $(-8+3\times 5)\times(5+4\times 8)\times(3\times 2-6)$.

4

Review Later

Two distinct even positive integers sum to 8. Determine the larger of these two integers.

5

Review Later

Suppose I had a bag of marbles, with many colors, and I wanted to find the probability that if I randomly chose a marble (with all marbles having the same chance of being chosen), I would get a red one or a green one. In order to do so, I would need to find _____ over _____.

- The total number of red and green marbles; the number of red marbles
- The number of red marbles; the number of green marbles
- The number of green marbles; the number of red marbles
- The total number of red and green marbles; the total number of marbles
- The total number of marbles; the number of red and green marbles

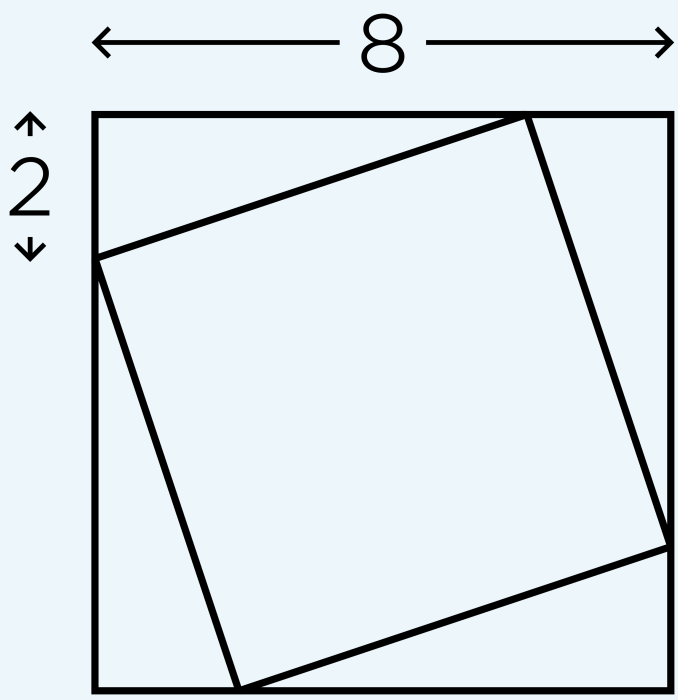
6

Review Later

Evaluate $29\times 29-33\times 25$.

7

Review Later



What is the area of the inner square, if the four right-angled triangles are the same?

8

Review Later

Albert, Beth, Carl, and Denise are trying to sit in a row of 4 seats in a movie theater.

How many ways can they sit down?

9

Review Later

Three faces of a rectangular box have areas 24, 20, and 30. What is the volume of the box?

10

Review Later

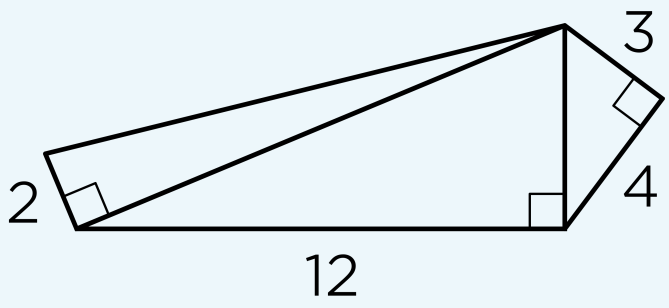
A rectangle with integer side lengths has an area of 12. What is the greatest possible perimeter of this rectangle?

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later



What is the area of this pentagon, created from three right triangles?

12

Review Later

How many prime numbers are divisible by 2?

- 0
- 1
- 2
- Infinitely many
- None of the above

13

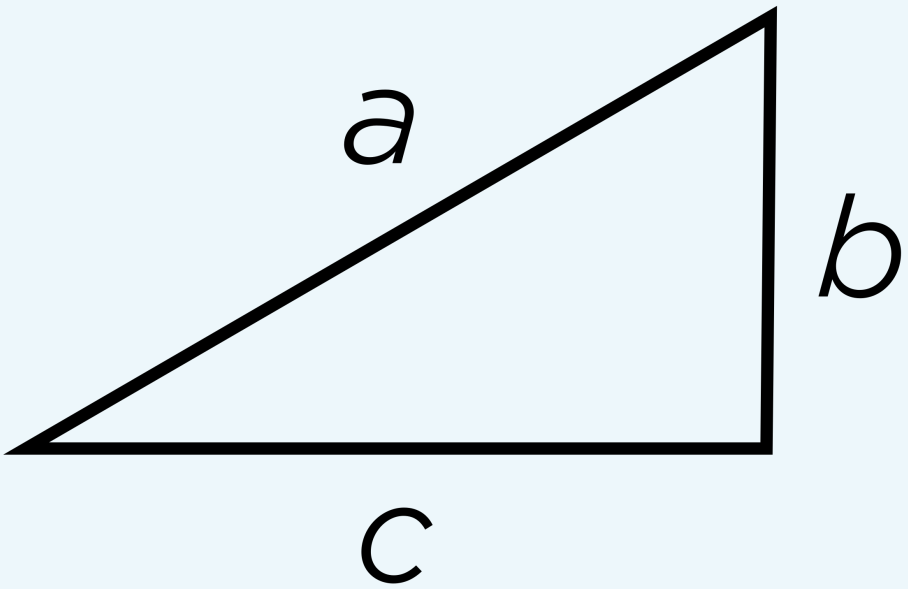
Review Later

Albert, Beth, Carl and Denise are trying to sit in a row of 4 seats in a movie theater. If Albert doesn't want to sit on either of the ends, how many ways can they sit down?

14

Review Later

Which of the following is true by the Pythagorean Theorem?



- $a^2 + b^2 = c^2$
- $a^2 + c^2 = b^2$

- $b^2 + c^2 = a^2$
- $c^2 + a^2 = ca$
- $a + b = c$

15

Review Later

How many 2-digit numbers do not contain the digit 1?

16

Review Later

How many 2-digit integers can we make such that the 2 digits are strictly increasing from left to right? For example, 12 and 36 would count, but 11 and 31 would not count.

17

Review Later

Suppose we want to find the number of 3 digit numbers which contain the digit 9 at least once. We can do this by subtracting _____ from _____ (called *complementary counting*).

- The total number of 3 digit numbers; the total number of 3 digit numbers which contain the digit 9 at least once.
- The total number of 3-digit numbers; the number of 3-digit numbers which don't contain the digit 9
- The total number of 3-digit numbers; the number of 3-digit numbers containing 2 of the same digits
- The total number of 2-digit numbers; the total number of 3-digit numbers
- The number of 3-digit numbers which don't contain the digit 9; the total number of 3-digit numbers

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

How many 3-digit integers have the property that the product of their digits is 30?

19

Review Later

Alina and Becky are playing a game where they flip a fair coin 3 times. Alina wins if the number of heads is even, and Becky wins otherwise. What is the percent chance that Alina wins, rounded to the nearest percent? (If the answer is 31.4 you would type in 31.)

20

Review Later

There are 6 empty chairs arranged in a circle. Three people arrive, and randomly choose places to sit, where all possibilities are equally likely. What is the percent chance that no two people sit next to each other, rounded to the nearest percent? (If the answer is 31.4%, you would type in 31.)

View Marked Questions

Finished

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